

ActInf OrgStream #004.1 ~ "Plural Publics", Shrey Jain

OrgStream | Shrey Jain | 55:58

hello and welcome it's March 31st 2023
we're in active inference or extreme
number 4.1 and here today with Trey Jane
one of the co-authors on the recent
peace plural Publix so Trey thank you
greatly for joining
and
we're going to start with a blank page
please feel free to introduce yourself
and the work and let's go from there
sounds great yeah thank you for having
me
um I'm really excited to share some of
the work that we've recently put out it
feels very timely but before we get into
it let me just give a brief intro as to
who I am
um so I'm sure I uh I currently work at
Microsoft research I sit in the special
projects division of Microsoft research
and collaborate frequently with our team
who is known as the plural technology
collaboratory
um where or PTC for short and we
intersect across various different Labs
at Microsoft as well as external
collaborators including the Harvard
software Center for ethics the Berkeley
decentralized research teams uh
decentralized like blockchain research
teams and various different other
partners that we have
um the reason why I'm on this podcast
today is in reference to some work that
I've been doing in collaboration with

Glenn Weil and debio siddarth which is
called plural Publix
um plural Public's uh
the way I'd like to introduce it is
starting a bit more with
what is a state of communication today
and why is this paper
especially important and to give a bit
of context here
when we reference the internet as a
communication tool we often reference it
to be the
the the most transformational tool that
we've seen in human communication and
that we reference it for for good
reasons as well we say that we're able
to communicate
basically with anyone across the world
who has an internet connection at almost
instantaneous speeds and that is
something that Humanity has never really
experienced in until the birth of the
internet and the applications that allow
us to communicate on top of it which is
very recent but the reasons under which
that has been actually quite harmful to
human communication is due to the
inability to preserve what's known as
context and context is this term that is
thrown around in academic language and
we want it to be very precise about its
definition in the paper and so the
definition that we give to context is
information that is common knowledge
amongst a set of people and so what I
mean by that is it's known to be known
it's known to it's known to be known
it's known it's known to be known as

known to be known to be known to known
it's like this like iterative Loop of
everyone knows that everyone knows
everyone knows everyone knows something
and to get a bit
to give a couple of examples of that is
when if I'm a doctor and I've read
medical textbooks
when I walk into a room with other
Physicians who I know have practiced
uh medicine in the country that I'm in
it is common knowledge that they have a
set of uh they have this dictionary that
we both share and this dictionary that
we share allows us to communicate
incredibly efficiently as an example
like if I had to walk into the operating
room and had to explain to someone what
some type of diagnosis meant every
single time we walked into that room
would be incredibly inefficient to
communicate and so because we have this
shared context but also because that
context is common knowledge it allows us
to communicate incredibly efficiently
and other examples we can see in our
life are things like inside jokes it
could be cultural milias it could be
religious Traditions it could be
National memories these are all things
that play a big role in the way that we
communicate with people around us and
some would argue that context actually
is the necessary components to
communication like if we didn't have
context even English itself would just
be a set of symbols and words that
didn't even make any sense to us right

like the only reason that you understand what I'm saying is because we have this shared context and symbols aren't just these like random pieces of Ink on paper but because we're in them were able to communicate and so that's really important like we really want to protect context and to go back to why I brought up the internet and its inability to preserve context is because when we share information online today is specifically on social platforms we don't preserve the context under which information was created and like as an example if I post something on Twitter about the conversation we're having on our podcast today maybe this is actually like not the best example let's say I post something on Twitter about a conversation I have at dinner with like um family members and it's like a joke that I thought was really funny and I wanted to post about well when I post that information on Twitter not only do my like family members friends or colleagues who are at the dinner table read that information but like the set of other people who read my Twitter content are interpreting that information with expectations dispositions and judgments that maybe that content wasn't initially intended to be communicated in and so context today is actually very hard to preserve like if you message something in a group chat it's actually very easy for someone to like screenshot and share

that into another chat or forward it to another chat or share something from Instagram into iMessage or from WhatsApp into like another platform and information just is Flowing everywhere and it's just like going crazy and so we can't preserve the Integrity of where it came from and so as a result we see a lot of negative repercussions of this people's expectations around a certain Community change when they don't interpret the information with maybe the way the community intended it to be communicated by or certain people who feel like they've become anxious or lonely as a result of this and we can reference some of the work by Dana Boyd and Helen nissenbaum which talks about the effects of context collapse in the teenage uh age group in particular and so anyway so so far like basically what we've what we've said is like the internet itself has this ability to not preserve context and that has been very harmful to human communication because context is critical to the ways in which we can communicate um well maybe we can go into now is like why we're under even more severe threat in the era of large models or large Foundation models in particular does that feel fine with you Daniel cool that was good so when I when I right now I guess let's talk about what are Foundation models what are genitive Foundation models and like

what all what is what is going on in the world right now with all of this so by now if you're listening to this podcast maybe there's a version of like GPT that's not only gpt4 GPT but it's even crazier if you're listening to this podcast within the recent time of it coming out you're already seeing the immense impact that chat GPT and gpt4 is having on probably you are people around you depending on the ecosystems under what you're living if you have been on an academic a bit been at an academic institution or a college campus in the past three months and you go to a library I guarantee you that around 50 of the students in the library have a split screen open with chat gpt and their actual coursework it's become a tool that is starting to impact the everyday work that everyone is doing um and so

this is this is both interesting as like a uh technological adoption uh framework but also very scary as like what are the repercussions that this is going to have on people and like the reason why it's interesting is like a technological adoption

um lens is because the adoption of chat gpt has been higher than any other social platform ever like it had more users and Twitter Tick Tock Instagram and all these other Platforms in the shortest amount of time possible but it's also really scary because the ability

that GPT like models or generative

Foundation models have
to decontextualize information even more
puts our information integrity at threat
and so like what do I mean by that
well first it's because these models are
able to create content in a very cheap
way what I mean by that you can create
bodies of essays you can create fake
profiles you can create fake Twitter
threads that
are seemingly real and what I mean by
that is they're like very persuasive
they look like they are coming from a
real human being
but the cost of creating that fake
content is so cheap
you're able to spread a bunch of
information on the internet in a
decontextualized way like you don't know
where it came from did it come from a
model or not can we prove that it came
from a model or not and so there's going
to be a lot of information polluting the
internet
that is going to cause people to look at
information in a way that they can no
longer Trust
and so what happens here is you start to
operate in a low trust environment and
what does that mean it means when you
look at information when you look at a
tweet when you look at an Instagram post
maybe even when you look at an email or
on a zoom call with someone
you can no longer trust that that is in
fact the authentic human version of
themselves
and so what does that lead to it leads

to a place in which
we are going to seek methods for
verification we're going to seek methods
under which people can actually
communicate with one another and have
confidence that that came from a human
but no longer do we have such confidence
because those tools today don't actually
exist and so

I guess to illustrate some further
examples is like how do we know where
that like a given

um output came from even a language
model if I'm even interacting with a
language model how do I know that that's
the language model that I'm even
interacting with like how do I know when
I go to chat GPT

website that it's actually going to be
chat GPT well today we have like some
actually cryptographic Primitives that
allow for that to instill some
confidence with like TLS certificates
but in the future as we maybe remove
like website domains and think about it
from social profiles

doing proof of humanity becomes a very
big problem and the impacts of this are
well discussed as it relates to deep
fakes and misinformation but we're not
focusing on that problem here we're
focusing more on the problem of
preserving context and communication and
for the reasons we described earlier we
want to be able to communicate
effectively with one another

um

so

Applause there and Daniel do you have any clarifying questions so far on anything well very interesting questions continue a little bit and then I'll definitely have a few things to ask okay so where we're at so far is like the internet already doesn't preserve context we're scared of it um ability to get exacerbated by large Foundation models and so I've sounded very pessimistic in this opening of this podcast but let me shed like a little bit of optimism here as well which is the internet has there's this like there's this element of the internet that's um like identity standards and identity protocols and the internet wasn't born with an identity protocol like the identity protocols today that the w3c is promoting like decentralized identifiers or verifiable credentials or actually very recent identity Primitives relative to like the birth of the internet and privacy in general has always been this thing where it's like we would love to have privacy and we like are actually going to build infrastructure and cryptography and New Primitives to enable such privacy because we recognize its importance but clearly given the way that we

communicate today it has not become something that is both a necessity for people to communicate as an example I still communicate today with like somewhat maybe not me but I'm saying like people in general with like weak passwords maybe not always using two-factor authentication and Twitter as a company may not even have hard requirements for me to authenticate that I'm in fact a human because guess what they want to promote virality and have more content on their platform and so they're going to make it really easy to sign up and and go and create an account and like the only thing you need to do to create an account today is have a Gmail account have a Gmail account you need I don't even it's not there's no strenuous requirements to prove your Humanity today to get these accounts and um

but what's really exciting here is like privacy has been always just like I guess nice thing that we could strive for but the the the harms that Foundation models pose to communication and the harms that Foundation models pose to the Integrity of information to the internet may actually make it a necessity for us to care about these privacy preserving tools because we won't be able to communicate otherwise and so like it's possible it's like actually almost exciting that the amount of noise the amount of damage that these

models can do
could create these channels of
communication on the internet that we've
actually never had before like maybe we
will be able to preserve context and
maybe we will be able to create common
knowledge when we communicate with each
other that we actually don't have today
because context is clearly collapsing
and context is clearly not being
preserved when we communicate and so
um this is kind of where we now can
start introducing plural publics and
like what exactly is plural publics and
like shedding a bit more light on the
optimism to how we build towards such a
future but before I go into Florida
Publix I want to give you time to also
ask any questions you have on the
problems framing
what is privacy I could easily imagine
one saying I was right with you on the
context and I'll be the one to control
that context or we'll have a context
asymmetry where I know a lot more about
you
so how do we really think about privacy
and in your mind what ties it so closely
with context
yeah so I want to reference here
Hellenism bombs work which
um she has this book titled privacy and
context
and in her work she argues how
what we what we really mean when we say
privacy is actually
um we mean contextual integrity and
let me explain what that means it's like

privacy today is this very

um

uh

how do I say it's like it's either

public or private that is like our view

of privacy today it's like things are

either open to viewing or closed to

viewing and there's this other framing

we maybe want to take which is that some

things that are open to some people may

be close to others and some things that

are close to some people maybe open to

others and that opening and closing is

highly dependent on the relationships

that we have with one another

and this is kind of like Helen Islam

articulates this much more clearly than

I am right now but I want to basically

try to frame it which is that

contextual Integrity is what we mean

when we say privacy privacy means that

something I feel confident in

communicating with you it may be private

in this context but I may not want to

share it with someone else and like how

we decide what is private and what is

not is going to depend on our social

norms it's going to depend on our prior

relationship and our context that we

have with one another and so

um another author who's written about

this

um

is krximo and in 1906 he has his paper

titled the sociology of secrecy and

secret societies and he writes a lot

about uh how

secrets are formed and secrets are

something that I find fascinating that I want to find a way to tie into here in a bit

um because

Secrets it's very possible that Secrets could be the only thing that we have to be able to authenticate ourselves in the digital world now in the sense that when I Who I Am to the public in terms of my like physical atoms and the way that I communicate through voice and the style in which I communicate given that I have even a very small public presence in my local communities but there exists content online about me could be enough to convey who I am to another audience in a very persuasive way and if anyone else has content about them as I think it's like a stat of like 80 of gen Z's I may be wrong about that stuff

is also able to be replicated in a very persuasive way now

but and this is where I get very excited our secrets can be the things that we can use to authenticate ourselves because those are the things that only the person who shares that experience with us or shares that context with us will be able to authenticate and what I mean by that here is like let's say Daniel me and you on this call assuming no one else was listening had a secret about a water bottle and this water bottle became a secret for some reason because either I don't know there was an interesting mixture that came in with something

about this water bottle became a secret
between us and so if I get on a zoom
call with you instead of doing captcha
are you a robot or like all these
standards of touring tests we have today
which I argue in the paper our touring
tests are going to be dependent on our
secrets

is that our secrets are going to be the
way we authenticate ourselves and this
is also exciting and in the sense of
like optimism which is that I think for
people who are able to and also I Can
Shed a bit of light on the fears I have
about this but like Because the Internet
may become unusable maybe we'll actually
live more in the physical world if we
can and we're able to and
develop more secrets such that we can
authenticate ourselves in the digital
world more like the more we just go on
like picnics and like have experiences
and like do things that cause them into
heartbreak but also immense happiness
will allow us to have more inside jokes
and it will allow us to have more
memories that we can say and allude to
when we want to authenticate ourselves
digitally

um but maybe yeah so that is like that
gives a sketch of kind of how I was
looking at your question
cool I'd love to hear a little more and
there's many active inference and
cognitive science threads that speak to
this so it's a cool to hear you go on a
little more about the plural publics
yeah so why don't we get into like

plural publics and I want to come back

to the secret stuff after as well so

um

first of all maybe I'll give this

framing of why I use the word plural and

then I'll get to the word public so

we'll get like very explicit here about

both of these terms so plural

um is in reference to this broader

um community of people working on what's

called plurality and if you want to read

more about plurality I want to reference

uh two people here Audrey Tong who is

the digital minister of Taiwan and Glenn

Weil who is my manager at Microsoft but

also more importantly one of the leading

figures uh embracing this movement of

plurality with Audrey Tom and they're

co-authoring a book together

um on plurality that you can read about

at plurality.net

um and they're doing it in this like

very interesting git format where

everyone can basically write pull

requests on the book and the chapters

have these like the way that the

chapters are developed they're done in

this like very cool deliberative

discussion way and anyways you can

Fanboy over that out of another time but

um

plurality

has many different axioms under which it

it is defined by but the Axiom that I

think sat well with me is how we get

um collaboration or coordination across

social differences and I think it's

really powerful to think about that in

in each world like what like how do you
coordinate how do you collaborate like
what are like the ways in which we do
that communication often depends on
sorry coordination and collaboration
depends on communication and
communication often depends on context
and there's like ways in which you can
build up how this all comes back to
plural publics

um but now let me get to the word Public
here which is

in reference to John Dewey John Dewey
um has this book called the public and
its problems and in 1927

um in the book he references what are
called emergent publics and emerging
publics are these groups of people who
come together to be able to deliberate
to ideate and Converse and discuss on
shared issues that they have with one
another

and find a way to resolve those issues
and it's the intersection of all these
different publics that come together to
form new knowledge and to form new
creation of this form new solutions to
problems that we're working on and so we
referenced this work plural publics as a
way

um to allude to John Dewey but before I
even get to plural politics actually I I
miss someone who is and I know I'm
referencing a lot of people in this in
this answer but it's really important to
to do so properly

um

and there's someone else here I want to

reference which is jcr LIC lighter jcr
look later was one of the internet
Pioneers who worked on what's called
arpanet is the precursor to the modern
day internet
and has this a series of work
that references uh a lot of the initial
principles and ideologies of emergent
publics and how the internet could be
this Global connectivity tool to
basically connect people and create a
more scalable form of emerging publics
online
and John Dewey's initial Vision sorry uh
jcr linklater's initial vision for the
internet and how it could abide by some
of the principles from John Dewey is
really exciting but due to the ways in
which the internet has been built and
the applications on top of it we've been
unable to get the form of public so
Dewey once imagined and what I mean by
that is like we're unable to coordinate
and build this collaboration across
social differences to the strength that
we may want to to solve
some of the most pressing problems we
have today and
the reason for that are manifold but one
of which is due to this inability to
preserve context online
and so plural Public's aims at taking
the vision that Dewey once imagined
along with the initial sketches that
arpanet took and try to reformulate the
problem uh into the understandings that
we've had with this initial
um I guess trial run of like the

internet and how we can like try to do
it again under a new framing in the
Perth of this new technology which is
generative Foundation models or AI
whatever you want to refer to as
and so we have a real opportunity here
to like actually build what you know our
um jcr look later wanted to do on
Dewey's vision and try again and we have
the opportunity to do it well now and
we're alluding to how plural publics now
can be that and so plural Public's
um in short are these digital spaces
where we can produce
common knowledge of context meaning
everyone knows that everyone knows
something in this given space and we can
both protect and establish that context
and so what I mean by that is
when I speak online I should both have
the confidence that I have the context
to communicate in that space and that
everyone who's listening to me has the
context to communicate in that space and
that I know that they have that context
and they know sorry give me one second
to take a sip of water yes
the secret water they know that oh I
think you were muted the secret water
bottle secret water bottle there we go
they know that I have that context but
more importantly I know that they know
that I know and I know that they know
that they are they know that I know that
they know and it's what this common
knowledge of context
that allows for such Rich forms of
communication to occur and the way the

way I usually imagine it is today we
have these social graphs in the world
which
um
leak information like someone might
share information
um outside of a given context and
information is very leaky what we want
to do is we want to build a social graph
that has extremely high information
integrity and what I mean by that is
both when I communicate I have extremely
high confidence in my communication
because I know everyone else has the
context to understand what it is that
I'm saying but secondly
um

I know that when I communicate that
information can't be leaked in a very
persuasive way so let me explain both of
these parts because now is how we get to
like the more interesting like how do we
build that what does that even mean you
seem like
visions that are not possible
so the first part I want to communicate
and have I want to be able to establish
I want to have confidence that everyone
understands what I'm saying how do we do
that and so in this paper we allude to
like five different technological
Primitives that we believe could be very
powerful in allowing us to achieve these
goals uh and I'll walk through how some
of these Technologies can help with
these two
um high level goals which is like both
establishing context and preserving

context so

I've I don't remember the exact order in which we presented it in but it was like I'll go through each of the five tools one of which is identity certificates so identity certificates today are

I guess I guess there's like certain subsets of communities that do use identity certificates and probably the web 3 Community is the most dominant form of like actually getting a comfortable understanding of the user experience and user interface for what that means and in particular instantiations of that look like um nfts where in order to enter a group chat I have to prove that I have a given certificate and that certificate today maybe through the issuance of a non-fungible token

um there's other examples of that in non-blockchain context in particular there exists uh verifiable credentials Anonymous credentials there's also um just very simple cryptography with um like PK like public key certificates that exist as well and so in general what I'm trying to allude to here is there's a bunch of cryptographic certificates that can be used to verifiably prove

someone has a certain property or attestation

the reason why that's important is let's say I want to communicate online maybe it might be helpful to prove that you have a set of properties whether it

be education whether it be through shared experiences like you were at the dinner last night and therefore you know what I'm talking about when I'm saying this thing or you went to that school for that year and took these courses and therefore you'll understand this set of terminology or you did XYZ and like if you know the limits of this become proving that you have the context basically to be in this space then we get to the second technology here which is distributed ledgers and so when you combine distributed ledgers with identity certificates you're able to create common knowledge of the context under which something may exist and so as an example sorry to rephrase it's like

um

when I communicate on a distributed Ledger and I prove that I have a certain attestation and you also prove that you have that attestation then I know that you have that attestation because there's a ledger that we've as globally agreed upon within this like set of people

that we both have that context to communicate

and there's actually some really interesting work out of Cornell from Joseph Halpern and Rafael Paz that proves as to why like distributed ledgers are actually

um

capable of creating what's called common knowledge or or actually in the paper

it's called approximate common knowledge
because of these properties of uh
blockchains or or non-blockchain type
distributed ledgers that are capable of
doing so
um so now we basically through these two
Primitives are able to both prove that
we have some context and both have the
confidence to know that everyone else
also has that context
now let's go to the second goal here
which is I want to make sure that when I
communicate it doesn't leave that
context so not only have we now
preserved it I will or sorry established
it I want to preserve the context so how
do we preserve context and so again I
guess I want to make a quick caveat here
when we wrote this paper
we had an idea of the tools that could
be useful
but by no means do we have the answers
and in fact everything I'm mentioning in
a couple of years could undermine the
exact goals that we're trying to achieve
but to our knowledge these are like the
the best tools to consider when when
thinking about these problems so I want
to make that quick caveat
um here but the first set of tools that
I want to reference for when we talk
about preserving context is what's
called deniable
or disappearing messages
and so I guess I'll give an example
today when I go to a coffee shop let's
say um Daniel and I go to a coffee shop
and there's no technology in the room

um we can make that assumption it's just us and we're just having a chat and when I communicate with you given that I know there's no technology and there's no phone recording in this conversation anything I say to you is going to be only verified by you that I said it meaning you can't go tell Jimmy some other person that Trey said this and Jimmy can prove that I said that because you have no proof that I even said it only you in this conversation know that I said

and so what this is what's known as deniable messaging only the receiver of that conversation

uh is capable of like proving the authenticity of the initial message from the sender and no third party is capable of proving the authenticity from the sender to the receiver

that is really powerful actually in information Integrity because if there exists a group chat on a distributed Ledger with a bunch of people who've proven through their identity certificates that they have some context then any message that is sent but sent in a way that's done it through a deniable message

only that group of people will be able to prove its authenticity

Matt is immensely powerful in a world where we don't believe anything and we're going to seek verification on every piece of content that exists in the world because we're going to assume due to the nature of AI and Foundation

models and generative content that everything is fake until proven otherwise and so given that given that I'm operating in this mindset that everything's fake and I'm seeking tools to verify the integrity and authenticity of information using deniable messaging techniques allows us to communicate in a way where only the people who can uh who can verify its authenticity are going to be able to do so and so now I can basically preserve the context on which information is sent and so like a prime example of that today is like if I try to forward a message to someone else through WhatsApp or whatever other platform you want to reference if that person who I share it with is not part of the designated verifiers or the people who should be able to authenticate they won't be able to and the specific primitive that I've been working on here in collaboration with some other researchers at the ethereum foundation um Enrico Potosi this is not like an ethereum Foundation project but just happens to be a project I'm working on with some people from there is building The Primitives to go out and actually Implement messaging techniques on ethereum with uh zero knowledge syllable tokens to send messages with designated of our signatures and we've created a component a JavaScript component that can be plugged into basically like any

application to allow people to do that
so these are tools that exist today
these are not just like hypothetical
cryptographic constructions these are
things that can be battle tested and out
in the wild today
um

so so I guess I've alluded now to like
various different technologies that
could be helpful in both preserving and
um protecting context and I want to now
just like bring it back to plural Publix
which is

if we're able to build a world where we
can have confidence in communication
because we can authenticate
the Integrity of that information and
ensure that it also doesn't leave the
context on which it was created
we're going to experience a form of
digital communication that we've never
been able to experience and I think is
actually quite beautiful

but if we're unable to do that in a
successful way

it becomes quite concerning actually and
we move into a world where there's
immense amount of distrust seeking for
verification

and scams to the degree in which we've
never seen I guess to make that very
explicit there was a recent scam that
happened on March 5th where
someone

um took the audio snippet of someone on
Tick Tock and they synthesized The Voice
from the girl on Tick Tock and called
that girl's Grandma as though they were

the voice of that person
and were able to use a various different
um techniques through conversation to
scam the grandmother
because the grandmother thought that
they were talking to the voice of a
loved one
when in fact it was just a scammer using
these different tools that already exist
today to persuasively scam someone
and so I guess what I'm saying here is
if you can't even trust the voice of
your loved ones if you can't even trust
the image of your loved ones you're
going to seek some set of tools to
communicate or else you won't
communicate at all
and this is very scary actually because
if you become unaware
of your ability to leak secrets about
yourself you become unaware of your
ability to just leak information about
yourself
I can only imagine a world where you now
are unable to access
your internet sure your like Financial
accounts and thereby your ability to
interact with a lot of the physical
world today
because you're unable to authenticate
yourself in the digital world because
you've become so easily reproducible
um
so this is like where plural publics
hopefully we start to create a set of
uh we hope to provide a framework in a
very high level framework but be useful
as individuals as co-authors of this

paper and people who are invested in
building Tech

to help people go out and like build
solutions to the types of problems we
allude to and with the Technologies we
sketch

I'll pause there and maybe we can open
it back up questions

awesome well many you're I think you're
muted then oh sorry only in the zoom
many interesting

topics and ways to go

I'll just bring in one active inference
idea and that's thinking through other
minds or ttom and in ttom

it's a lens to think about communication
where individuals don't just have a
theory of mind it's not just that they
have some belief about whether the
entity that they're conversing with like
has an awareness which is kind of an
interesting philosophical point but
really more pragmatically thinking
through other Minds points to the
emulation

of a conversions perspective

and that being one of the critical
aspects of communication overall not
like some Edge recursive case like you
brought up it's not enough just that you
know it's that you know that they know
and that's what allows the low friction
and high fluency high bandwidth high
integrity communication is not needing
to ask which of these English words do
you understand or which one of these
authors are you familiar with and so
it's um really interesting to see how

what might be seen as kind of this
esoteric or Edge case where we're
emulating or as if emulating each
other's perspective that's actually the
Crux of communicating across difference
of which we are always doing
just different kinds of differences that
we might have a certain belief about
which ones are not crossable or not
agreed yeah and I'll just put this paper
in the chat here
um a pre-print that came out by some
faculty at Stanford around how theory of
Mind tasks have emerged
through large language models and
they're capable of completing theory of
Mind tasks and so
um thought that might be an interesting
read for for the audience as well
and and one related
um area especially that you're I think
pointing to with
um this work
first we care about the cognitive
security and the well-being of ourselves
and of humans
however increasingly these conversations
also can be understood as cognitive
yes cognitive models that have
differential degrees of agencies or
capacities in the world
but it's what merits what philosopher
Daniel Dennett called the intentional
stance which is like interacting across
that interface that Turing test
interface instead of treating it like a
test past no pass
you can treat it from a game theory

perspective in the intentional stance
way like I don't know whether it's a
human underneath that Mechanical Turk or
whether it's a crowdsourced or this kind
of model or that kind of model
however there are ways that we can
approach it that are basically the best
way to approach it in a given setting
yeah and that intentional stance
we've gotten by far and well
with bootstrapping on local context like
taking intentional stance towards an
embodied human in a given party is
really easy but when it's a chat room
it's not and so it's almost like when
the comic book character runs off the
cliff and still runs for a little bit
like we're on that bootstrapped inertia
of context I have to say I totally
believe that it's really you
but again
how rapidly should that belief be
updated downwards in the coming several
months
maybe all the way to the floor
and that is going to be the issue
it's very interesting you say that
because
um part of me like
when I when I read about
um information asymmetry in
relationships
there's always been this sense of like
asymmetry in the sense that like you
don't know with with 100 confidence what
I'm thinking about even if you're my
partner
but like that's been okay like we've

actually operated with some asymmetry in
our relationships like we still make
progress we still live in like the NFL
game still plays at night like
everything still happens
but like I think there's actually an
interesting relationship here between
information asymmetry and context
collapse which is uh as we've
increasingly communicated online the
asymmetry that exists between people's
expectations of the world or the way
that people themselves move or operate
has grown to like a delta or whatever
you want to call that asymmetry
difference to be larger and larger and
larger to the point where like as you're
talking about here is like with large
Foundation models maybe become so big
that we can't even communicate anymore
and like what's worked for Humanity of
like asymmetry in the physical world
like no longer becomes functional in the
digital
um and that's actually quite scary to
think about too so anyways yeah one
really logistical example of that is
time zones I mean in person you're
always talking to somebody in the same
time zone weather condition
but the way in which in a video
conversation
it could be sunset it could be sunrise
and
the kinds of practices that emerged even
rapidly over the last several years to
help at least reduce uncertainty which
is the imperative in active inference

not to maximize reward but to bound or minimize our surprise these practices can be understood as bounding surprise on these important axes of context like oh cool I haven't seen you in this meeting how long have you been in this organization or oh what time is it where you at or how's the weather outside those help really rapidly dampen our uncertainty and that's what enables effective policy selection with uncertainty about the mapping between observations in the world or uncertainty about the consequences of action in the world yeah you can't act yeah yeah no I I really like that Framing and it's like yeah it's interesting because I think there's almost like a negative connotation towards like people on Zoom calls asking like how's the weather outside like it's a waste of time to like ground the conversation in that like first two minutes put like maybe there is actually something more important and subtle in that that like maybe I haven't appreciated it as much but maybe I should be more aware of as like yeah like it's snowing and raining outside where I am and how does that affect the way I communicate as compared to someone who's looking outside and seeing a sunny Beach and like the small things and the context on which we're actually like sitting in a room and maybe like I have an issue in my apartment out here right now and like that's gonna affect the way I

communicate here anyways

yeah

and and one other uh area or type of context and one that people have started

to model within active inference

framework is narrative context because

I'm thinking about those two people

face-to-face talking

so they're seeing entirely different

Landscapes maybe one is facing them out

and one's facing away so it's clearly

not enough to have shared context at the

level of exactly what one sees or hears

but there is some level of course

graining where

the people will expect and prefer to

have shared context and whether it's a

micro narrative like something very

situational hey in the next 30 minutes

we're here to do this or that

or whether it's a more broad or even

open-ended Adventure type Journey

having tools to increase our bandwidth

of communication around rhetorical and

narrative context is going to set the

stage for so much more to happen but if

we're scrambling with the pixels while

two people have different ideas about

what genre of movie they're in ones that

a horror movie one's in a rom-com then

you can't

um just pile sand onto that kind of a

foundation whereas in fact have a better

overarching ways for communities

to self-determine and build resilient

systems than we may

new kinds of flourishing that haven't

been enabled in web one two or three

yeah I couldn't agree more it's um I think we're adding a very like the the feeling I get right now with everything coming out is like we are at a point where like we have an actual opportunity to fix communication in the way we like never have been able to like we can get the properties of physical communication of like being in the same room and having this like context of one another but doing so in a way that we can also benefit from like Global instantaneous communication that maybe we've never actually had um and that's really cool that's really exciting and I think that we're at a point now where we're seeing claims from some of the world's smartest people um who are building these Technologies saying that like oh Foundation models could be as revolutionary as the wheel or electricity not even the internet but like as revolutionaries like some of these even more drastically important technologies that we should take this very seriously if we meet those words the way that they're being said um and treat it as also an opportunity to build around it and and hopefully improve human communication but yeah so I think like that so like I think that's like wraps up like kind of the plural public streaming and like it was very like nice to actually walk it through in the way that we did here it feels like

we went through like at a good pace and
feels nice yeah

I guess just in our

Landing of the plane how do you and your
colleagues move forward and how can
people be included in their own ways in
how things will develop

yeah I guess a couple things so

um

one is I as an individual and like
pretty accessible if you want to reach
out to me and like find a way to
collaborate on things whether it's like
me and my affiliations with Microsoft

Pro or even easier for me as an
individual wanting to just promote and
support any projects that I can I will
do my best to do so secondly is that we
at Microsoft research have launched the
plural technology collaboratory and you
can read more about I'll just put a link
in the chat here the the work I'm
leading at Microsoft research on plural
publics in particular

um and you can read more about that
effort

um on this blog post we made about Pearl
Publix

um and there's a bunch of different open
source projects we have on behalf of
Microsoft research that people can
contribute to at a technical level as
well whether that be the

um

uh go for get projects which you can
read more about or the you proof project
these are all projects that you can look
at and lastly is like I think

um I do think that there's a lot of reading and things that people can do to like also get excited about themselves it's like try to if you can read the paper or read versions of the reference papers we reference in the in that paper I think that you will change the way you even perceive sometimes the things that are going on with the large Foundation models movement right now or whatever you want to call it because you may see it as either pessimism depending on how you interact with these tools or great optimism but you I hope that when you read this work you also see it with a great opportunity to be a contributor to it whether you as an individual who can is technically competent or you as an individual who can just embrace the way that you use these tools and take great care with how you and the people around you use it and I think there's like various different ways in which plural Public's involvement could take place and also my Twitter I'm pretty active on Twitter um you can DM me there as well Jane eth is my Twitter handle so you thoughts and angles that we could just sort of bring in near the end um there are opportunities for technical contributions like you've brought up and maybe even what will Dawn is the social competencies being able to engage in perspective swaps and then prompt the kinds of

Technologies

that would scaffold

those Communications

yeah it may be less about going into the

for Loop and more about really having

the conversations and interactions that

help shape inclusive conversations

and there isn't a script or an algorithm

for that

so as the ground shifts beneath our feet

with what kinds of projects have what

kind of technical hurdles for even small

groups

that continues to evolve that so that's

um one very interesting angle

for sure for sure I couldn't agree more

and then one angle to kind of connect to

active inference and our area here

is

the kinds of information ecosystems that

are

arguably always existed but

today coming into our conscious design

and control

these ecosystems of communication have

all kinds of cognitive entities they

have true natural humans they have daos

they have organizations they have

augmented entities people who are using

typos in spellcheck all the way on

through people who are doing social

engineering scams like you brought up

and there are also increasingly purely

digital entities in these information

ecosystems or ecosystems of shared

intelligence and so from our community

we see that in important technology will

be composable and flexible graphical

architectures statistical models that help us describe communication patterns amongst and within different plural publics so that instead of just saying well it's happened to turn out how it's happened to turn out and here's how I'm feeling arbitrarily about whether we should

go this way or go that way

we can have something like a design

Suite that allows uh visualized and executable conversation so people can say actually I'd like if the small teams were four people instead of three how could we do that how can we re-render our organization to do that differently and what might be some of the consequences and so again without going into technical details a policy that you similarly abided by hopefully in the coming years it'll be easier than ever to make those kinds of statistical analyzes of cognitive entities more available

yeah I couldn't agree more and I'm very supportive of the work active inference is doing and

um yeah I know I I I can't express the the need for those types of research problems but also building the tools to enable that kind of inference to make people actually take action on it too in a way that doesn't detriment the decision making that we have already great well

what are your closing words or thoughts or exhortations

maybe like the last thing I'll say and I

just like can't reiterate it anymore I

was like

this is like a really cool time to like

be working or thinking and having fun

sparring with people

in this space and like

if you just engage him in the slightest

way and do it in a way with like a bit

of optimism and like not too much of

like

long-termism when the World's Gonna

collapse

at least that's how I try to go about it

um it feels like a very

it feels refreshing a bit because

there's like if you can band together a

group of people who want to do this

properly I think that it's it's a lot of

opportunity so find ways to like engage

in a way with optimism and I think that

we can build some really cool stuff

together if so

amazing well the body doesn't have a

long-term plan it also only has a

sequence of short-term plans

and so that is absolutely the question

and the positive approach to approach it

expecting and preferring survival and

flourishing and then enacting short-term

actions so that we can really make it

work long term

the and like the we also may not have 10

years like we may only have like a year

or two to like get this done so it's

also there is like urgency I guess in

that as well

um and so it's like

yeah like I just say start like get

going if you're not already going or
like just ramp it up because we like
this is not something to like this is a
short-term need we need to start acting
on

thank you Trey for joining

thank you everybody checks out the paper
and future developments of the work so
thank you Daniel in the active inference
yeah see you till next time thank you